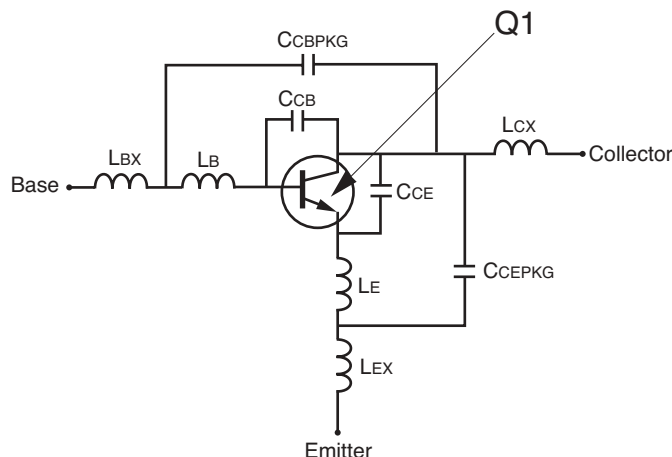


SCHEMATIC



BJT NONLINEAR MODEL PARAMETERS ⁽¹⁾

Parameters	Q1	Parameters	Q1
IS	3.84e-16	MJC	0.64
BF	124.9	XCJC	0
NF	1.05	CJS	0
VAF	11.9	VJS	0.75
IKF	0.027	MJS	0
ISE	1.0e-14	FC	0.5
NE	2.17	TF	8.7e-12
BR	1	XTF	18
NR	1.05	VTF	19.1
VAR	Infinity	ITF	0.082
IKR	Infinity	PTF	0
ISC	0	TR	0.635e-9
NC	2	EG	1.11
RE	0.6	XTB	0
RB	17.9	XTI	3
RBM	1.02	KF	0
IRB	4.01e-4	AF	1
RC	10.5		
CJE	0.358e-12		
VJE	0.71		
MJE	0.5		
CJC	0.162e-12		
VJC	0.79		

(1) Gummel-Poon Model

Life Support Applications

These NEC products are not intended for use in life support devices, appliances, or systems where the malfunction of these products can reasonably be expected to result in personal injury. The customers of CEL using or selling these products for use in such applications do so at their own risk and agree to fully indemnify CEL for all damages resulting from such improper use or sale.

CEL California Eastern Laboratories, Your source for NEC RF, Microwave, Optoelectronic, and Fiber Optic Semiconductor Devices.
4590 Patrick Henry Drive • Santa Clara, CA 95054-1817 • (408) 988-3500 • FAX (408) 988-0279 • www.cel.com

DATA SUBJECT TO CHANGE WITHOUT NOTICE

UNITS

Parameter	Units
time	seconds
capacitance	farads
inductance	henries
resistance	ohms
voltage	volts
current	amps

ADDITIONAL PARAMETERS

Parameters	68019
CCB	0.08e-12
CCE	0.08e-12
LB	0.72e-9
LE	0.76e-9
CCBPKG	0.17e-12
CCEPKG	0.21e-12
LBX	0.19e-9
LCX	0.19e-9
LEX	0.19e-9

MODEL RANGE

Frequency: 0.05 to 3.0 GHz
Bias: $V_{CE} = 1\text{ V to }6\text{ V}$, $I_C = 1\text{ mA to }15\text{ mA}$
Date: 8/03